

Now let us take the case of the area of the segment APB of a circle with centre O and radius r (see Fig. 11.4). You can see that :

$$\begin{aligned}\text{Area of the segment APB} &= \text{Area of the sector OAPB} - \text{Area of } \triangle OAB \\ &= \frac{\theta}{360} \times \pi r^2 - \text{area of } \triangle OAB\end{aligned}$$

Note : From Fig. 11.3 and Fig. 11.4 respectively, you can observe that:

Area of the major sector OAQB = πr^2 – Area of the minor sector OAPB
and Area of major segment AQB = πr^2 – Area of the minor segment APB

Let us now take some examples to understand these concepts (or results).

Example 1 : Find the area of the sector of a circle with radius 4 cm and of angle 30° . Also, find the area of the corresponding major sector (Use $\pi = 3.14$).

Solution : Given sector is OAPB (see Fig. 11.5).

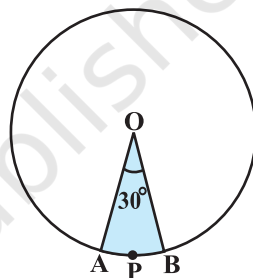


Fig. 11.5

$$\begin{aligned}\text{Area of the sector} &= \frac{\theta}{360} \times \pi r^2 \\ &= \frac{30}{360} \times 3.14 \times 4 \times 4 \text{ cm}^2 \\ &= \frac{12.56}{3} \text{ cm}^2 = 4.19 \text{ cm}^2 \text{ (approx.)}\end{aligned}$$

Area of the corresponding major sector

$$\begin{aligned}&= \pi r^2 - \text{area of sector OAPB} \\ &= (3.14 \times 16 - 4.19) \text{ cm}^2 \\ &= 46.05 \text{ cm}^2 = 46.1 \text{ cm}^2 \text{ (approx.)}\end{aligned}$$

Alternatively, area of the major sector = $\frac{(360 - \theta)}{360} \times \pi r^2$

$$\begin{aligned}&= \left(\frac{360 - 30}{360} \right) \times 3.14 \times 16 \text{ cm}^2 \\ &= \frac{330}{360} \times 3.14 \times 16 \text{ cm}^2 = 46.05 \text{ cm}^2 \\ &= 46.1 \text{ cm}^2 \text{ (approx.)}\end{aligned}$$

Example 2 : Find the area of the segment AYB shown in Fig. 11.6, if radius of the circle is 21 cm and

$$\angle AOB = 120^\circ. \text{ (Use } \pi = \frac{22}{7} \text{)}$$

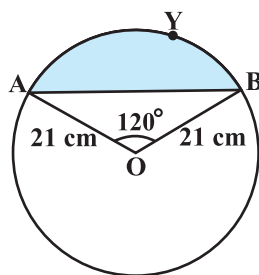


Fig. 11.6

Solution : Area of the segment AYB

$$= \text{Area of sector OAYB} - \text{Area of } \triangle OAB \quad (1)$$

$$\text{Now, area of the sector OAYB} = \frac{120}{360} \times \frac{22}{7} \times 21 \times 21 \text{ cm}^2 = 462 \text{ cm}^2 \quad (2)$$

For finding the area of $\triangle OAB$, draw $OM \perp AB$ as shown in Fig. 11.7.

Note that $OA = OB$. Therefore, by RHS congruence, $\triangle AMO \cong \triangle BMO$.

So, M is the mid-point of AB and $\angle AOM = \angle BOM = \frac{1}{2} \times 120^\circ = 60^\circ$.

Let $OM = x$ cm

So, from $\triangle OMA$, $\frac{OM}{OA} = \cos 60^\circ$

$$\text{or, } \frac{x}{21} = \frac{1}{2} \quad \left(\cos 60^\circ = \frac{1}{2} \right)$$

$$\text{or, } x = \frac{21}{2}$$

$$\text{So, } OM = \frac{21}{2} \text{ cm}$$

$$\text{Also, } \frac{AM}{OA} = \sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\text{So, } AM = \frac{21\sqrt{3}}{2} \text{ cm}$$

$$\text{Therefore, } AB = 2 AM = \frac{2 \times 21\sqrt{3}}{2} \text{ cm} = 21\sqrt{3} \text{ cm}$$

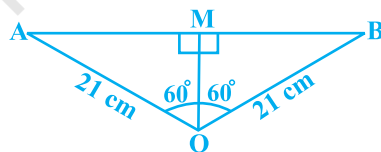


Fig. 11.7